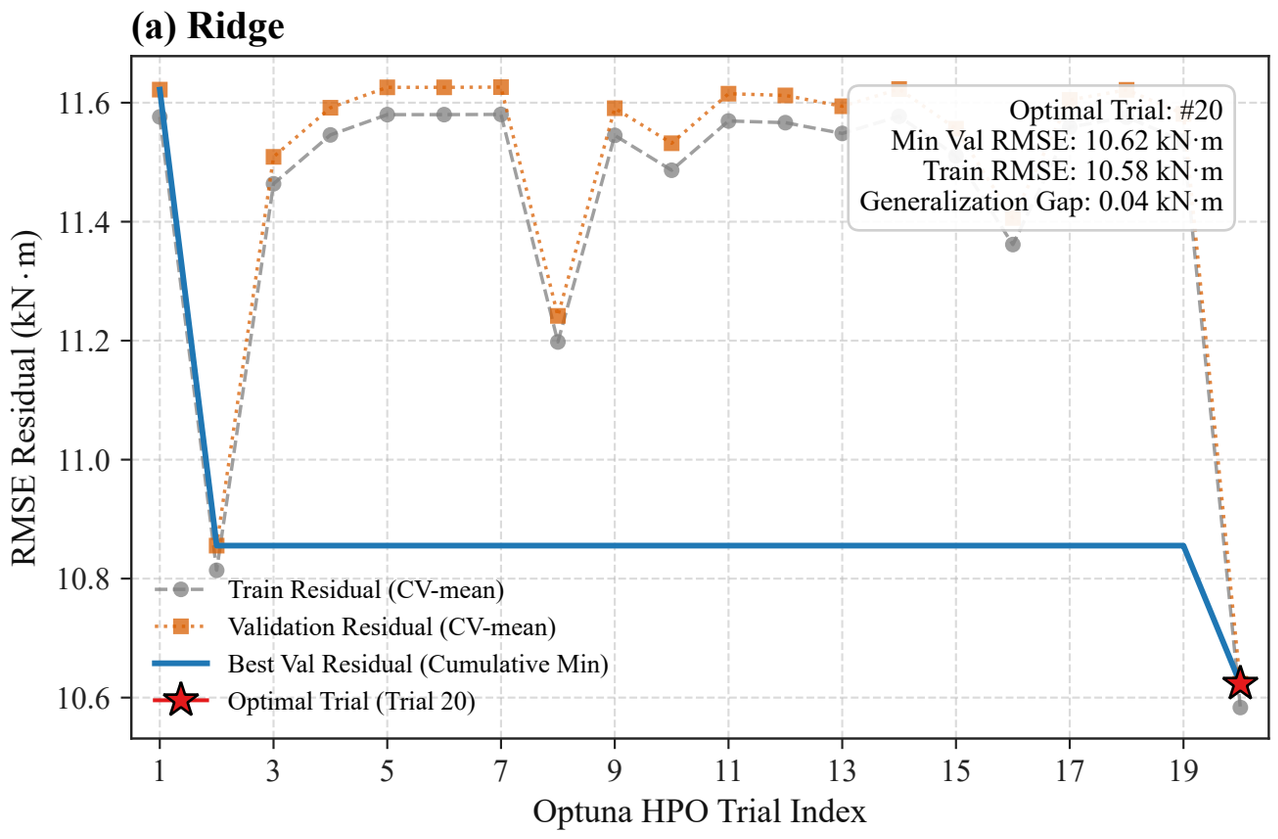
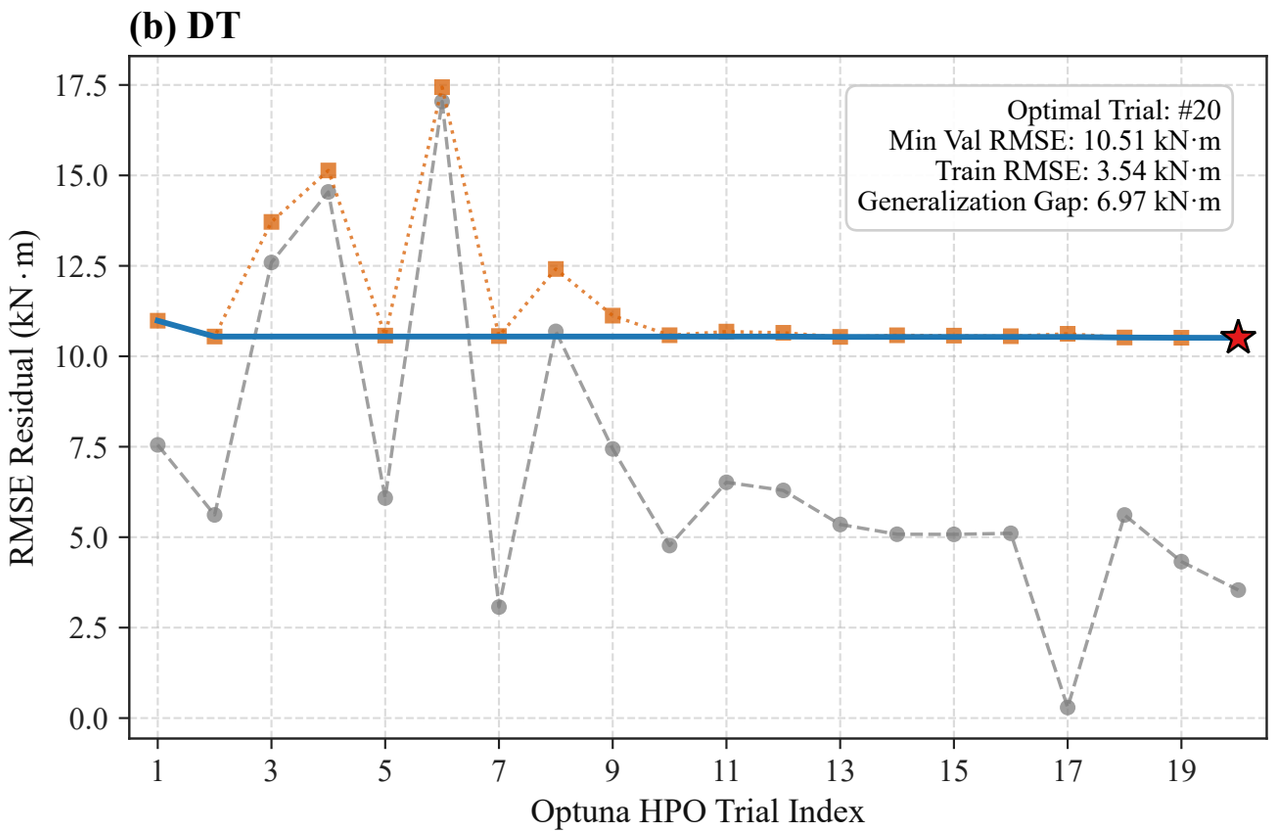
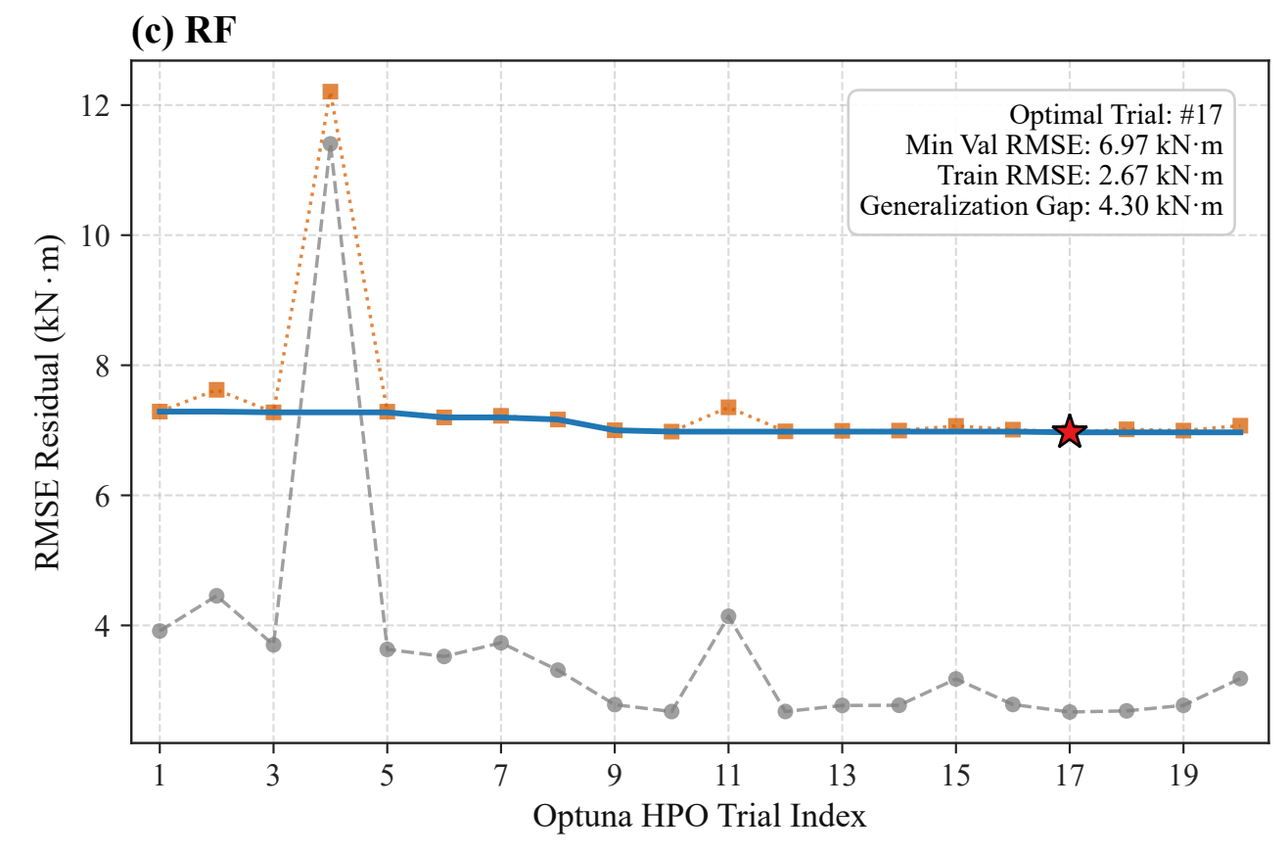
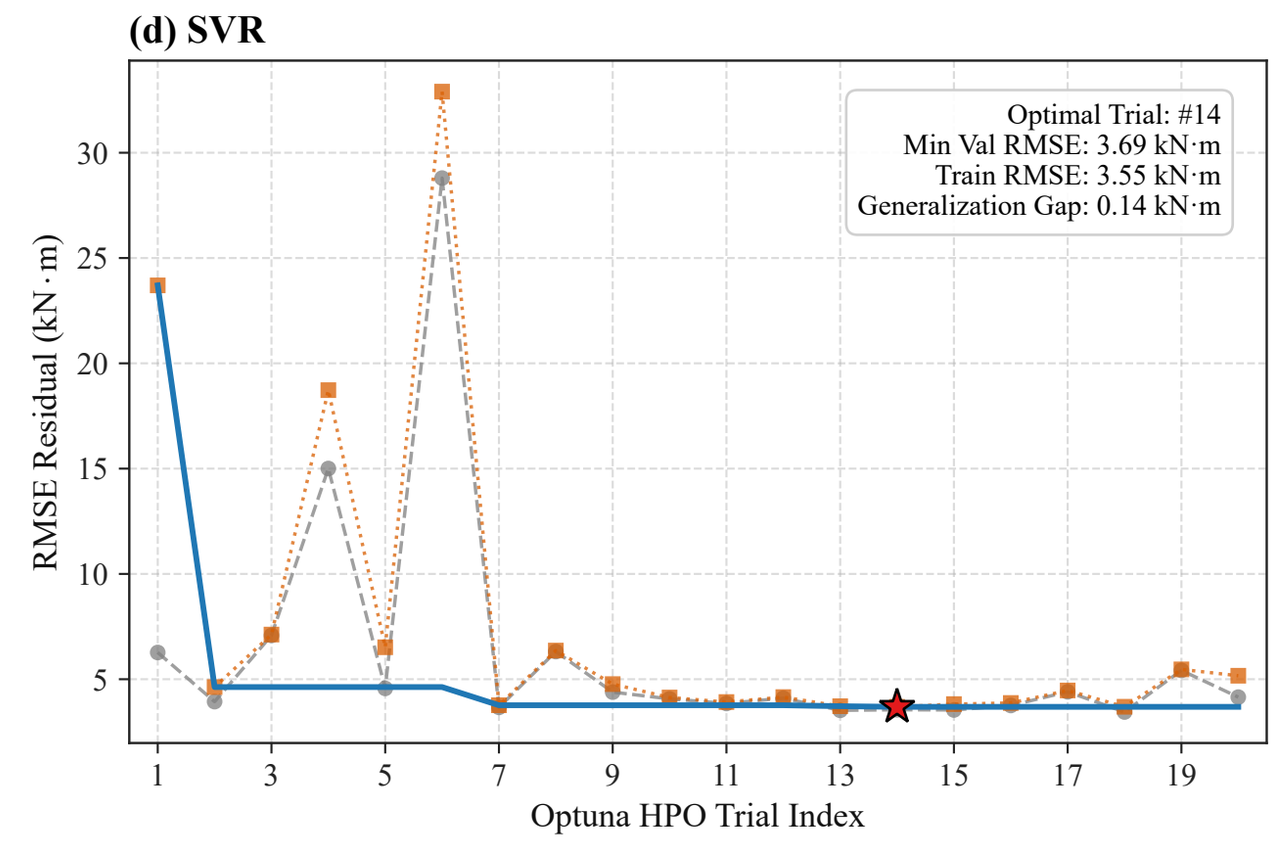
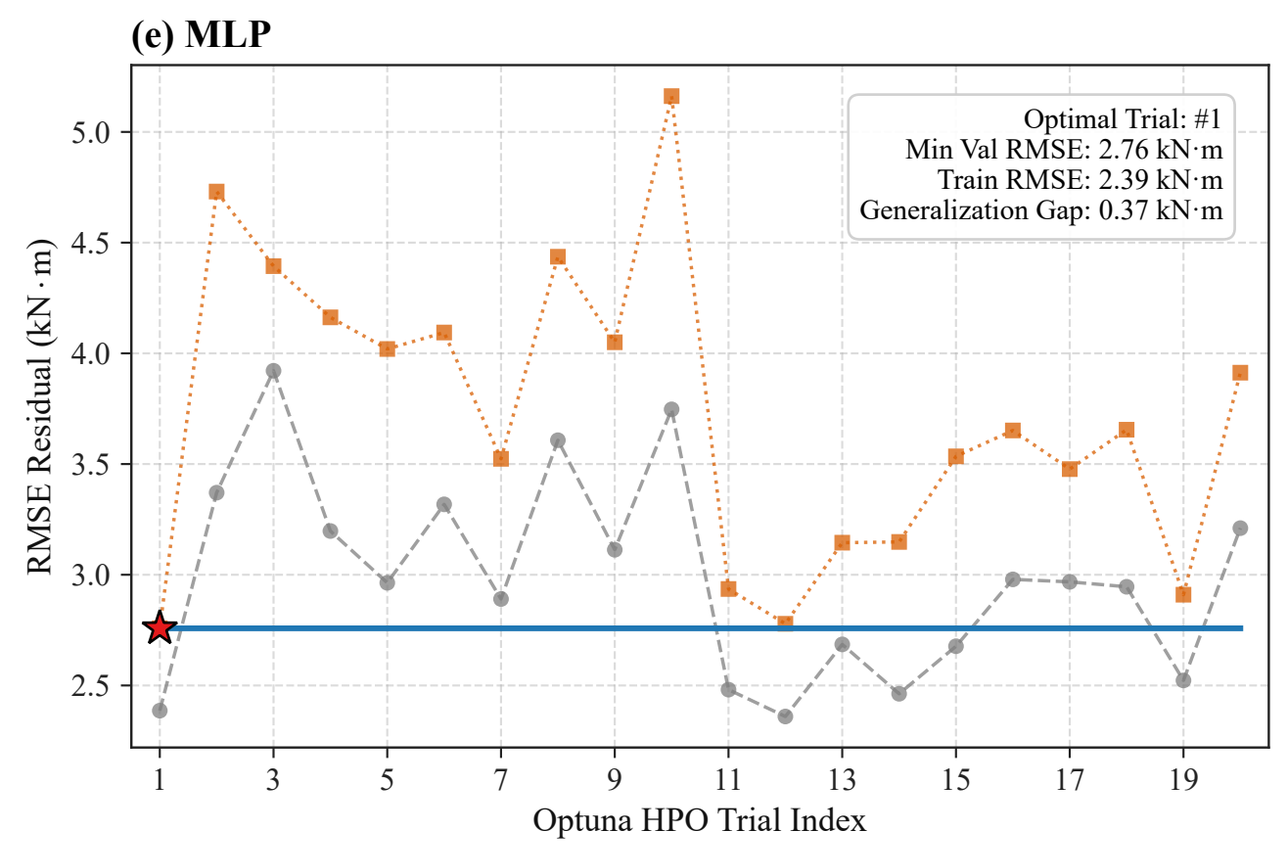
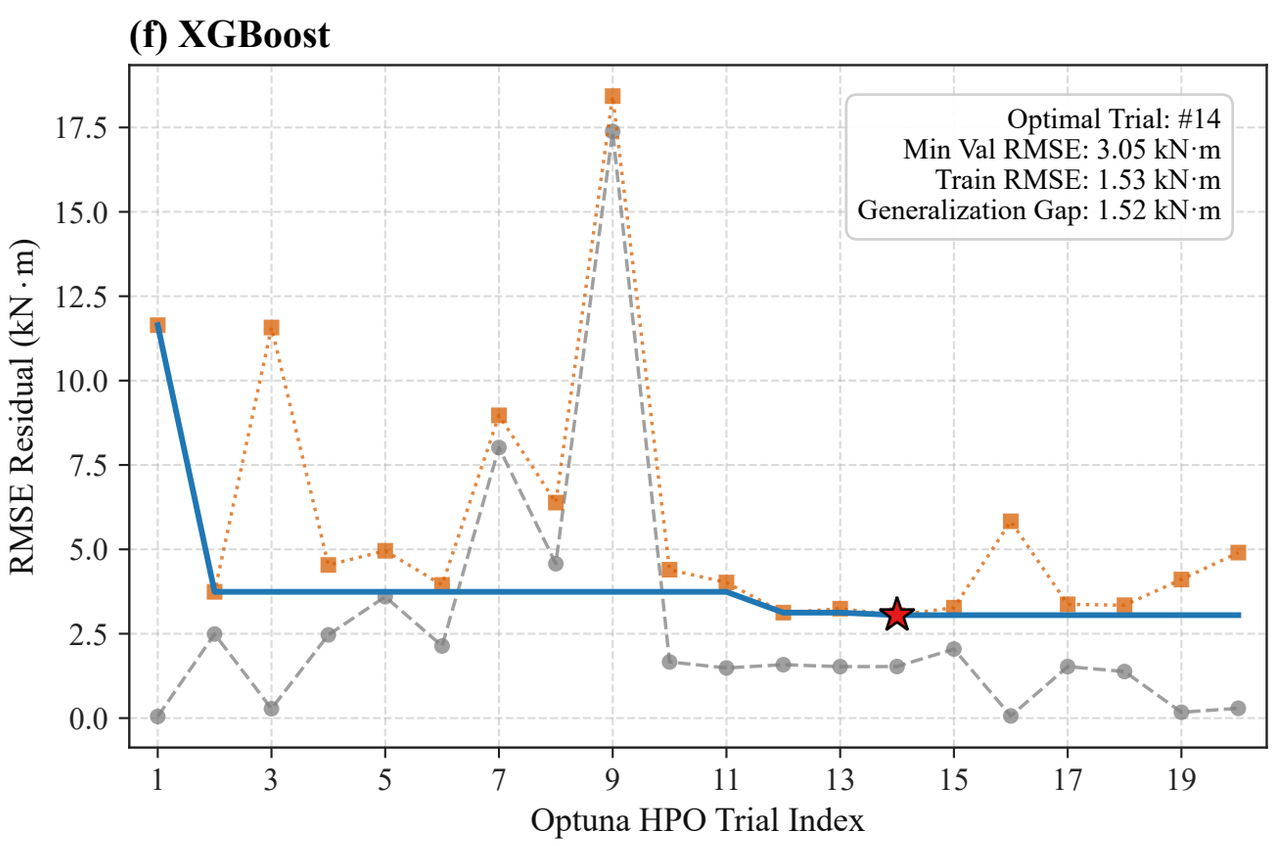

(a) Ridge
Optimal Trial: #20
Min Val RMSE: 10.62 kN·m
Train RMSE: 10.58 kN·m
Generalization Gap: 0.04 kN·m
Train Residual (CV-mean)
Validation Residual (CV-mean)
Best Val Residual (Cumulative Min)
Optimal Trial (Trial 20)
RMSE Residual (kN · m)
Optuna HPO Trial Index

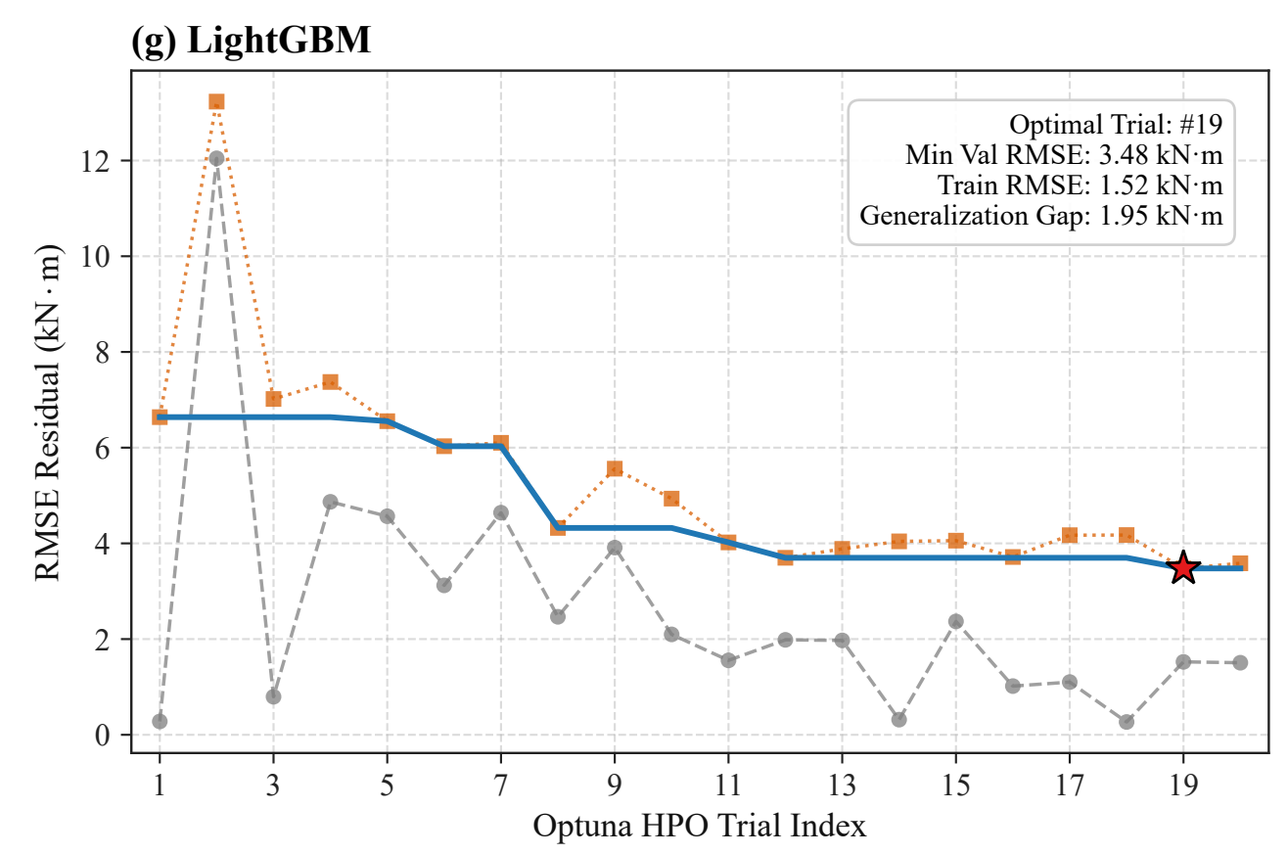

(b) DT
Optimal Trial: #20
Min Val RMSE: 10.51 kN·m
Train RMSE: 3.54 kN·m
Generalization Gap: 6.97 kN·m
RMSE Residual (kN · m)
Optuna HPO Trial Index

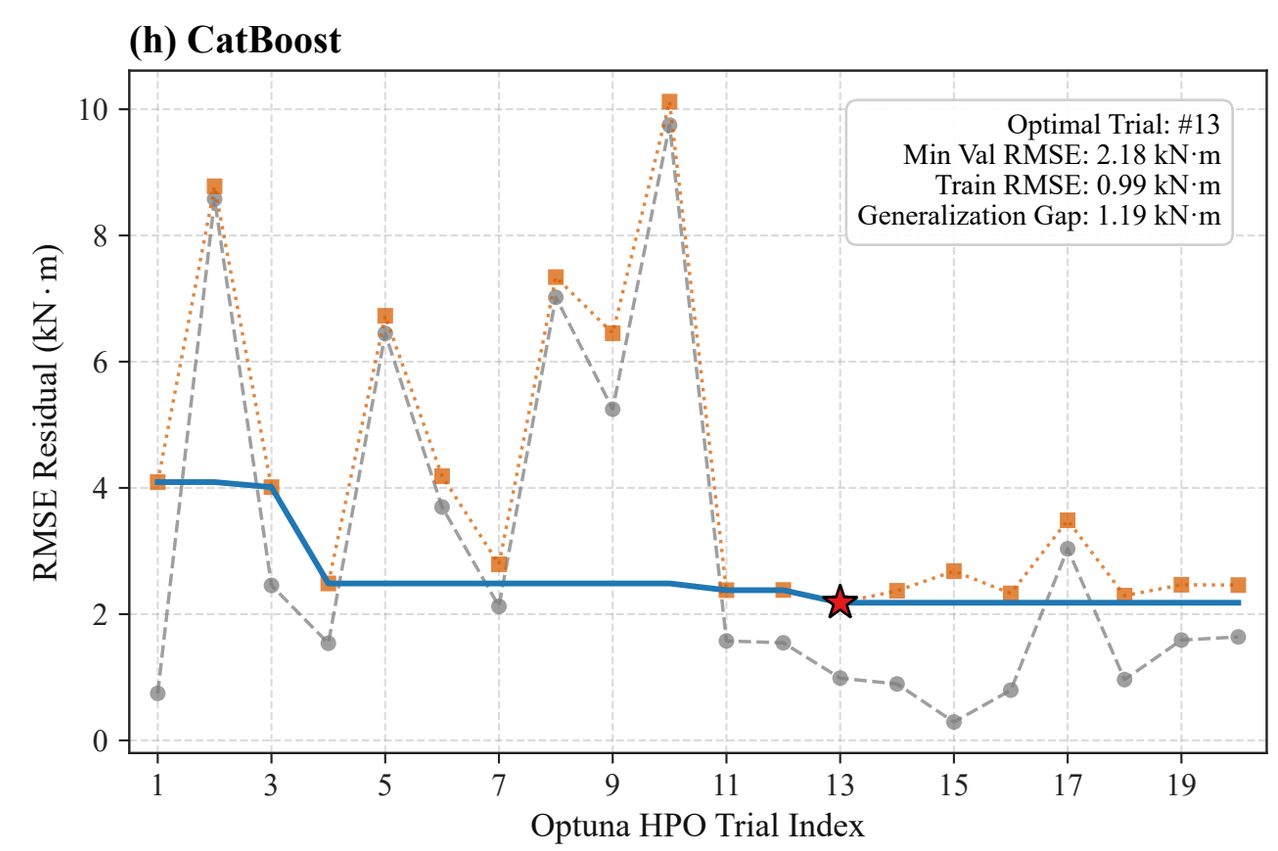

(c) RF
Optimal Trial: #17
Min Val RMSE: 6.97 kN·m
Train RMSE: 2.67 kN·m
Generalization Gap: 4.30 kN·m
RMSE Residual (kN · m)
Optuna HPO Trial Index


(d) SVR
Optimal Trial: #14
Min Val RMSE: 3.69 kN·m
Train RMSE: 3.55 kN·m
Generalization Gap: 0.14 kN·m
RMSE Residual (kN · m)
Optuna HPO Trial Index


(e) MLP
Optimal Trial: #1
Min Val RMSE: 2.76 kN·m
Train RMSE: 2.39 kN·m
Generalization Gap: 0.37 kN·m
RMSE Residual (kN · m)
Optuna HPO Trial Index


(f) XGBoost
Optimal Trial: #14
Min Val RMSE: 3.05 kN·m
Train RMSE: 1.53 kN·m
Generalization Gap: 1.52 kN·m
RMSE Residual (kN · m)
Optuna HPO Trial Index


(g) LightGBM
Optimal Trial: #19
Min Val RMSE: 3.48 kN·m
Train RMSE: 1.52 kN·m
Generalization Gap: 1.95 kN·m
RMSE Residual (kN · m)
Optuna HPO Trial Index


(h) CatBoost
Optimal Trial: #13
Min Val RMSE: 2.18 kN·m
Train RMSE: 0.99 kN·m
Generalization Gap: 1.19 kN·m
RMSE Residual (kN · m)
Optuna HPO Trial Index